



LINUX HARDENING TECHNIQUES

PRESENTED BY : **Badal Kamble**

EXAM SEAT NO : **ZI32681**

COLLEGE : **Dayanand College of Commerce, Latur**

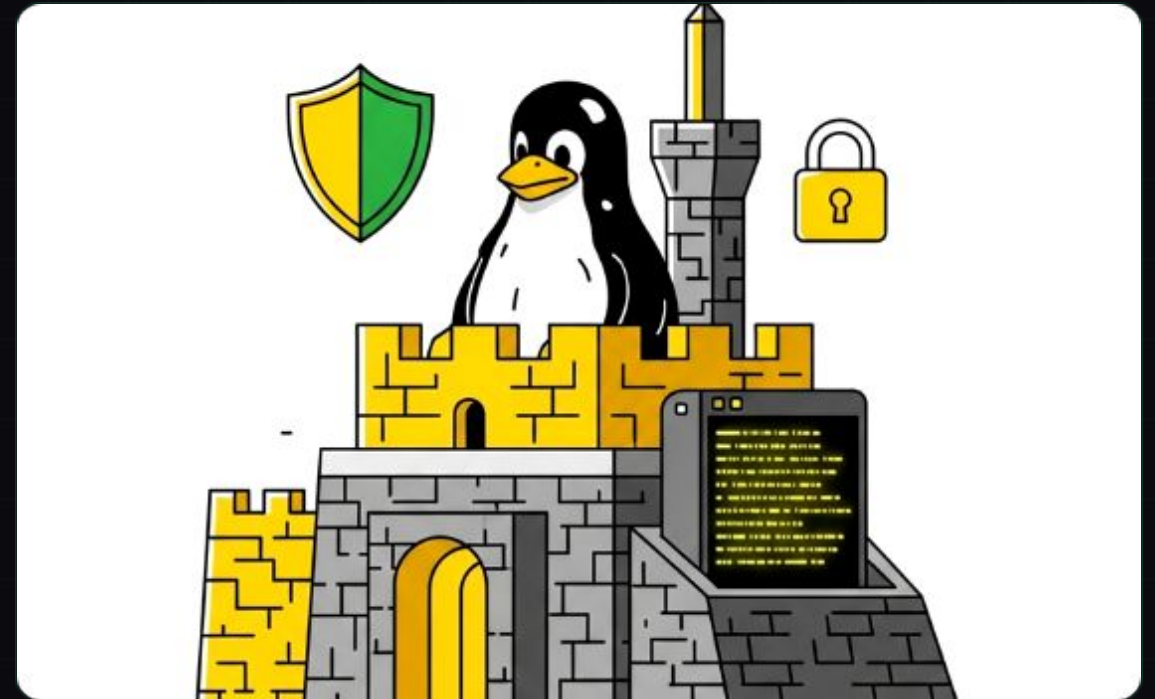
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01_INTRODUCTION

WHAT IS LINUX HARDENING?

Linux Hardening is the process of securing a Linux system by reducing vulnerabilities, controlling access, and applying strict security configurations.

- > Protects from cyberattacks & malware.
- > Prevents unauthorized access & data breaches.
- > Ensures stability & reliability for real-world use.



02_WHY_USE_HARDENING?



- > **Protect Servers:** Defend against active hacking attempts.
- > **Reduce Attack Surface:** Disable unnecessary features.
- > **Access Control:** Prevent unauthorized login & privilege misuse.
- > **Data Protection:** Stop sensitive data leaks.
- > **Compliance:** Meet industry security standards.
- > **Damage Control:** Minimize impact even if a breach occurs.

03_TECHNIQUES (1/2)



SYSTEM UPDATES & PATCHING

Keep OS and software up-to-date to fix known security bugs immediately.

Automated Defense:

 **Live Kernel Patching:** Use tools like KernelCare or Canonical Livepatch to update kernels without rebooting.



> Example: 60% of Linux attacks occur due to unpatched systems.



SSH HARDENING

- > Change default SSH port.
- > Use key-based authentication.
- > Disable password login.
- > Install Fail2ban.

SSH is the primary remote entry point. Default configurations are often insecure.

> Example: Fail2ban blocks thousands of brute-force attempts weekly.

04_TECHNIQUES (3/4)

FIREWALL CONFIGURATION

Use tools like UFW or FirewallD. Allow only necessary ports and block everything else.



> Example: Blocking unused ports reduces attack risk by > 50%.

USER & PASSWORD HARDENING

- > Enforce strong passwords.
- > Disable unused accounts.
- > Restrict 'sudo' access.
- > Disable Root Login.

> Example: Disabling root login prevents direct brute-force attacks.

05_TECHNIQUES (5/6)

📁 FILE & PERMISSION HARDENING

Use least privilege principles. Secure mount options:

- > /tmp → noexec
- > /var/tmp → nosuid

📁 Filesystem Permissions

Restrict where scripts can execute to prevent malware from running.

> Example: 'noexec' stops malware scripts from running in temp folders.

🚫 DISABLE UNNECESSARY SERVICES

Turn off unused background services to reduce the attack surface.



> Example: Disabling CUPS/Bluetooth reduces attack chances.

06_TECHNIQUES (7/8)

🛡️ User Identity & Access

🌐 Strong Authentication

Enforce strong password policies using libpam-pwquality. Require minimum length (12+ chars), complexity, and rotation for privileged accounts.

📱 Multi-Factor Auth (MFA)

Implement MFA for all SSH and Sudo access. Tools like Google Authenticator or Duo provide a critical second layer of defense against credential theft.

👁️ LOGGING & MONITORING

Enable and monitor system logs using Journald, ELK, or Splunk.



> Global Data: 90% of security breaches are detected through logs.

07_GLOBAL-REAL-WORLD_DATA

90%

of cloud servers worldwide are powered by Linux.

20k+

Linux vulnerabilities reported yearly (CVE Database).

160%

Increase in brute-force attacks in the last 2 years.

40%

of companies attacked due to poor configuration.

CONCLUSION



Linux Hardening is essential for keeping systems secure, stable, and protected from cyber threats.

By implementing Patching, SSH Security, Firewalls, and Monitoring, we make Linux systems significantly harder to attack.

"A hardened system reduces risk, protects data, and ensures long-term security."

"Security is a process, not a product."

// Thank You